# CUSTOMER SEGMENTATION AND CLUSTER ANALYSIS ON RETAIL CUSTOMER BEHAVIOR

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GitHub repository: github.com/aaherigangopadhyaystats-code

## ABSTRACT

Customer heterogeneity is important for effective retail marketing. This project analyses 200 mall customer profiles (n = 200) using unsupervised learning to identify groups based on demographic characteristics, income, and spending behaviour. Exploratory analysis examined univariate distributions, correlations, and the effects of feature scaling. K-Means, agglomerative hierarchical clustering, and density-based approaches were considered. The Elbow Method applied to inertia values indicated a five-cluster solution in the two-dimensional space defined by annual income and spending score. The resulting groups included high-income high-spenders, high-income low-spenders, and other distinct spending profiles. Z-score standardization was then used to support four-dimensional clustering across age, annual income, spending score, and gender. The findings illustrate how clustering can provide interpretable customer segments that may inform tailored marketing strategies.

Keywords: customer segmentation; K-Means; hierarchical clustering; unsupervised learning; retail analytics; feature scaling

## 1. PROJECT OVERVIEW AND RATIONALE

Retail businesses may struggle to engage customers effectively when the customer base is treated as a single homogeneous group. Grouping shoppers with similar income levels and spending habits can help businesses design targeted campaigns aligned with different customer profiles.

The project addresses three practical questions:

- Do income and spending habits naturally form identifiable customer segments?
- What number of clusters provides a useful balance between mathematical compactness and business interpretability?
- How do feature scaling and categorical attributes such as gender affect cluster boundaries when moving from two-dimensional to multivariate clustering?

## 2. EXPLORATORY DATA ANALYSIS

The Mall Customers dataset contains 200 customer profiles and five primary variables. Customer ID was excluded from clustering because it functions as an identifier rather than a meaningful behavioral feature.

| Variable | Description |
|---|---|
| Gender | Categorical; 112 female customers (56%) and 88 male customers (44%). |
| Age | 18–70 years; mean 38.85; standard deviation 13.97. |
| Annual income | Reported range $15k–$137k; mean $60.56k. |
| Spending score | Scale from 1 to 100; mean 50.20. |
| Customer ID | Unique identifier, removed before clustering. |

The dataset contained 200 rows with no missing values. Exploratory analysis reported a slightly higher average spending score among female customers (51.53) than male customers (48.51), although average income was slightly higher among male customers ($62.23k versus $59.25k). Age had a moderate negative correlation with spending score ($r = -0.327$). Annual income and spending score had almost no direct linear correlation ($r = 0.0099$), suggesting that income alone is insufficient to characterize spending behaviour.

## 3. PREPROCESSING AND FEATURE PIPELINE

- **Feature cleanup:** Customer ID was removed because it contains no substantive customer-behaviour signal.
- **One-hot encoding:** Gender was converted into a binary feature (Gender-Male).

- **Feature standardization:** Z-score scaling was applied for multivariate clustering so that variables measured on different scales contributed more comparably to distance calculations.

The standardization formula was:

X_scaled = (X – Mean) / Standard Deviation

## 4. CLUSTERING EXPERIMENTS AND FINDINGS

### 4.1 Univariate K-Means: Annual Income

K-Means clustering was first applied to annual income alone with K = 3. This separated customers into income tiers but did not capture differences in how customers spend.

| Cluster profile | Mean income | Size |
|---|---|---|
| Low income | $33.49k | 74 |
| Moderate income | $69.75k | 104 |
| High income | $108.18k | 22 |

Reported inertia for this three-cluster model was 23,589.31.

### 4.2 Bivariate K-Means: Annual Income and Spending Score

K-Means was then applied to annual income and spending score. The Elbow Method compared within-cluster sum of squares (inertia) for K values from 1 to 10. The reported values were:

| K | Inertia |
|---|---|
| 1 | 269,981.3 |
| 2 | 181,363.5 |
| 3 | 106,147.1 |
| 4 | 73,679.7 |
| 5 | 44,448.5 |
| 6 | 37,233.7 |

The elbow was identified at K = 5. The resulting five customer segments were:

| Segment | Income profile | Spending score | Mean age |
|---|---|---|---|
| High-income, high-spending | $86.54k | 82.13 | 32.7 |
| High-income, low-spending | $88.20k | 17.11 | 41.1 |
| Moderate income, moderate spending | $55.30k | 49.52 | 42.7 |
| Low-income, high-spending | $25.73k | 79.36 | 25.3 |
| Low-income, low-spending | $26.30k | 20.91 | 45.2 |

The segments distinguish customers whose income and spending scores do not align in a single linear pattern. In particular, high-income customers appear in both high- and low-spending groups, supporting the use of segmentation rather than income alone for profiling customer behaviour.

### 4.3 Multivariate Clustering and Alternative Algorithms

The analysis was extended to four standardized features: age, annual income, spending score, and the binary gender feature. The reported elbow pattern again suggested approximately five groups. Agglomerative hierarchical clustering with Ward’s linkage and K = 5 produced boundaries described in the original report as closely resembling the two-dimensional segments.

## 5. TAKEAWAYS AND FUTURE WORK

- **Distinct spending profiles:** Annual income alone did not capture spending behaviour. The two-dimensional K-Means model separated high earners into high- and low-spending groups.
- **Potential marketing applications:** The high-income, high-spending segment may be relevant for premium campaigns, while the low-income, high-spending segment may respond to trend-oriented promotions. These are possible applications inferred from the cluster profiles, not experimentally validated campaign outcomes.
- **Future work:** Density-based clustering such as DBSCAN and silhouette scores were identified as possible next steps for examining border points and quantifying cluster separation.

## CONCLUSION

The analysis identifies five interpretable customer segments in the income–spending space and reports a similar broad grouping pattern after adding age and gender to the feature set. The findings illustrate the value of exploratory analysis, feature standardization, and comparing clustering approaches when developing customer profiles. Because the analysis is unsupervised and descriptive, the segments should be treated as useful summaries of this dataset rather than proof of future purchasing behaviour.